AL-RIYADA COLLEGE FOR HEALTH SCIENCES · MEDICAL HRMS

# Installation and Setup Guide

Install Medical HRMS and the Jameah (Ministry) data views on a Frappe server, set them up for the college, and check that everything works.

| | |
|---|---|
| FOR | The server administrator who installs and updates the system, and the system owner who sets it up (HR Manager / System Manager). |
| COVERS | Medical HRMS app ( `medical_hrms` ) including the Jameah views, branch `develop` . |
| VERSION | 23 October 2026 |
| BUILT BY | Abdul Hakeem (Medical HRMS and Jameah lead) |
| SEE ALSO | User Guide (how HR, managers, employees and testers use the system). |

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# 1 What gets installed

Medical HRMS is one Frappe app, `medical_hrms`. It adds the college's HR processes on top of the standard Frappe HR (HRMS) app, and the Jameah views the Ministry's data agent reads. The standard apps are not modified.

| Part | What it gives the college |
|---|---|
| **HR Portal** | One page for HR: requests waiting for a decision, leave setup check and sick leave, recruitment, Ministry data, academic presence ( `/app/hr-dashboard` ). |
| **Employee Portal** | Self-service for every employee: 11 request types, the leave flow (plan, apply, return), balances and monthly limits ( `/app/employee-dashboard` ). |
| **Requests** | Permission, Remote Work, Salary Certificate, Overtime pre-approval, Training, Children Medical Allowance, Company Car, Employee Data Update, Leave Plan, Leave Return, Contract Renewal (HR only), Medical Licence. |
| **Leave** | Editable starter leave types and policies (annual 21 / 30 days), sick leave in 3 pay stages with its own sick year, monthly half-day and work-from-home limits. |
| **Recruitment** | Job Application Links, the public application form, review, and Create Employee (with login and, for academic roles, Instructor / Academic Advisor / Head of Department records). |
| **Ministry codes** | 9,867 official Ministry lookup codes, loaded on every migrate, used on Employee and applicant fields. |
| **Jameah views** | 23 read-only database views in the Ministry's format: Employees 5, Faculty 8, Students 10, with a Data Check page, a Code Map and an independent verification. |
| **Academic presence** | Faculty and proctor schedules, students on campus per hour, faculty after 4 PM. |

# 2 Before you start

- A Frappe **v15** bench with these apps installed on the site: `frappe`, `erpnext`, `hrms`, `education`.
- MariaDB, Redis and Node (as for any Frappe bench). Medical HRMS needs no extra packages.
- Workers running for the **short, default and long** queues, and the **scheduler** enabled. The Jameah academic figures are rebuilt every hour on the long queue.
- Access to the GitHub repository `phpdevpro/edu-alriyada-hrm`. The branch to install is `develop`.
- A backup of the site before you install or update: `bench --site <site> backup --with-files`.

> **The education app must be current**
>
> The Ministry GPA and credit hours use the college's official calculation that comes with the latest `education` app (master branch). Update it on the server in the same deployment.

# 3 Install on a new site

## 3.1 Install the app

Run these commands as the bench user, in the bench folder. Replace `<site>` with the site name.

```
cd ~/frappe-bench
bench get-app git@github.com:phpdevpro/edu-alriyada-hrm.git --branch develop
bench --site <site> install-app medical_hrms
bench --site <site> migrate
bench build --app medical_hrms
bench --site <site> clear-cache
bench restart
```

The app folder must end up as `apps/medical_hrms`, and `sites/apps.txt` must list each app on its own line (see Troubleshooting).

> **Read the migrate output**
> It must finish without a `Traceback`. The last part rebuilds the Jameah academic figures and takes a few minutes. If it prints a line starting with *Kept (they hold data ...)*, send that line to the developer.

## 3.2 What migrate sets up

Every `migrate` runs the app's setup steps, so a new site and an updated site end up the same. Nothing has to be run by hand.

| Step | What it does |
|---|---|
| **Ministry codes** | Loads or updates the official Ministry lookup codes (Jameah Ministry Code). |
| **Ministry fields** | Adds the Ministry and Jameah fields and tables to Employee and Instructor (identity, nationality, job rank, qualifications, experience, ...). |
| **Jameah** | Creates the Jameah doctypes and settings, refreshes the Code Map, creates the 23 views and rebuilds the academic figures. |
| **HR permissions** | Gives HR roles the access their pages need (for example read on Company, Holiday List, Instructor). |
| **Employee Self Service** | Creates the employees' workspace. |
| **Leave starter records** | Creates the leave types (annual, Sick Leave 1 / 2 / 3, unpaid) and draft policies if they are missing. Existing records and HR's changes are kept. |
| **Monthly limits, presence** | Adds the HR Settings fields for monthly half-day / work-from-home limits and for academic presence. |
| **Patches** | One-time data changes listed in `patches.txt` (for example the sick-leave renaming). |

**Do not run the old setup scripts**

Earlier notes listed scripts such as `configure_hrm_request_governance`, `quick_grant_hr_permissions` and `apply_hr_workspace_layout`. Do not run them: the doctypes now carry their own permissions, and `configure_hrm_request_governance` would replace them.

# 4 Update an existing server

1. Back up the site: `bench --site <site> backup --with-files`.
2. Get the new code:

```
cd ~/frappe-bench/apps/medical_hrms
git fetch origin
git checkout develop
git pull --ff-only origin develop
```

3. Update the `education` app to its current master in the same deployment (see section 2).
4. Apply it:

```
cd ~/frappe-bench
bench --site <site> migrate
bench build --app medical_hrms
bench --site <site> clear-cache
bench restart
```

5. Run the checks in section 5.

> **Timing**
>
> The hourly Jameah refresh makes pages slower for a few minutes while it runs. A migrate never fails because of it any more (only one refresh runs at a time), but schedule updates outside busy hours.

# 5 Check the installation

| Check | How | Expected |
|---|---|---|
| **Scheduler and workers** | `bench --site <site> doctor` | Scheduler enabled; workers for short, default and long. |
| **Ministry views** | `bench --site <site> execute medical_hrms.jameah.verify.run` | Every view **OK** (23). |
| **Error Log** | Open `/app/error-log`, newest first. | No new errors since the update. |
| **HR login** | Log in as an HR Manager. | The HR Portal opens. |
| **Employee login** | Log in as an employee whose user is linked to an Employee record. | The Employee Portal opens with their name. |
| **System e-mail** | Email Account *Notifications* → send a test e-mail. | The e-mail arrives. |
| **Logo** | Open `/files/rchs-logo.png`. | The college logo shows. |

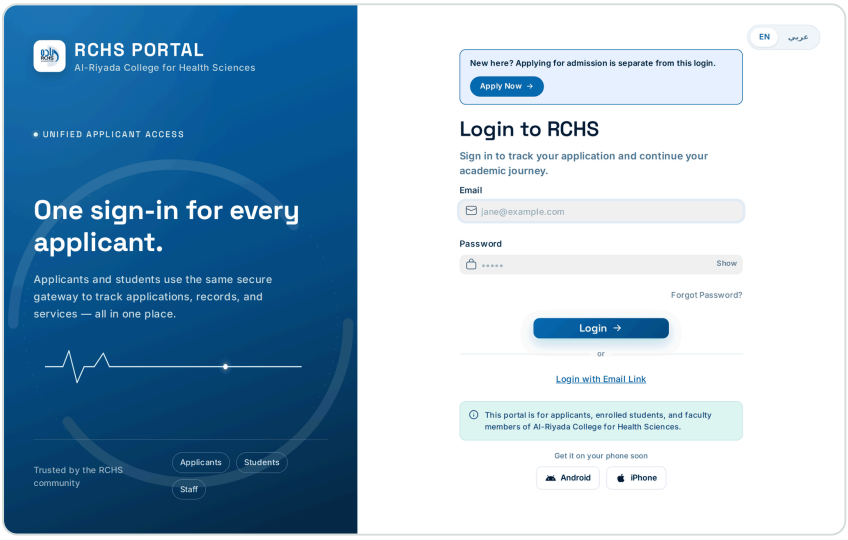



**Figure 1.** The login page. HR users land on the HR Portal and employees on the Employee Portal.

# 6 First-time setup

Done once by the system owner (HR Manager or System Manager) after the first installation, and checked at the start of every leave year.

## 6.1 Users and roles

Every person who logs in needs a **User**. Employees also need their User linked on their **Employee** record (field **User ID**); this gives them the Employee role and the Employee Portal.

| Role | Who has it | What it allows |
|---|---|---|
| **HR Manager** | Head of HR, HR system owner | Everything in the HR Portal, including the staff data sheets, starting a sick leave year, Jameah Settings and the Code Map. |
| **HR User** | HR officers | Process requests, employees, leave and recruitment. Read-only on Jameah Settings and the Code Map; cannot use the staff data sheets or start a sick leave year. |
| **Employee** | Every member of staff | The Employee Portal: own requests, own leave, own records only. |
| **Leave Approver** | Managers named as leave approver | Approve the leave of the employees they approve for. |
| **Department Head** | Heads of department | Approve training, travel and expense claims above the department-head limit. |
| **Finance Manager / Accounts Manager** | Finance | The finance step on money requests and amounts above the finance limit. |
| **Education Manager** | Academic Affairs | Jameah Data Check, Jameah Settings and the Code Map. |
| **System Manager** | IT | Everything (administration). |

Approval limits by amount (checked when a request is approved):

| Request | Department Head | HR Manager | Finance |
|---|---|---|---|
| **Training** | above 10,000 | above 30,000 | above 50,000 |
| **Travel** | above 15,000 | — | above 50,000 |
| **Expense claim** | above 5,000 | — | above 20,000 |
| **Company car** | — | — | always (final approval) |
| **Children medical allowance** | at most 3 children per academic year; at most 30,000 approved per child | | |

## 6.2 Company and e-mail

- Check the **Company** record of the college (Al Riyada College): default holiday list (section 6.3) and the company e-mail domain.

- Academic hires must use their official college e-mail ( `@alriyada.edu.sa` ) as Company Email; the Create Employee dialog asks for it.
- Open Email Account **Notifications**, enter its password again and save, then send a test e-mail. Until this works the system cannot send e-mails.

## 6.3 Leave, in this order

The HR Portal has a **Leave** section with a **Leave setup check** per company and the yearly setup steps in order. Work through the steps until every line of the check is green.

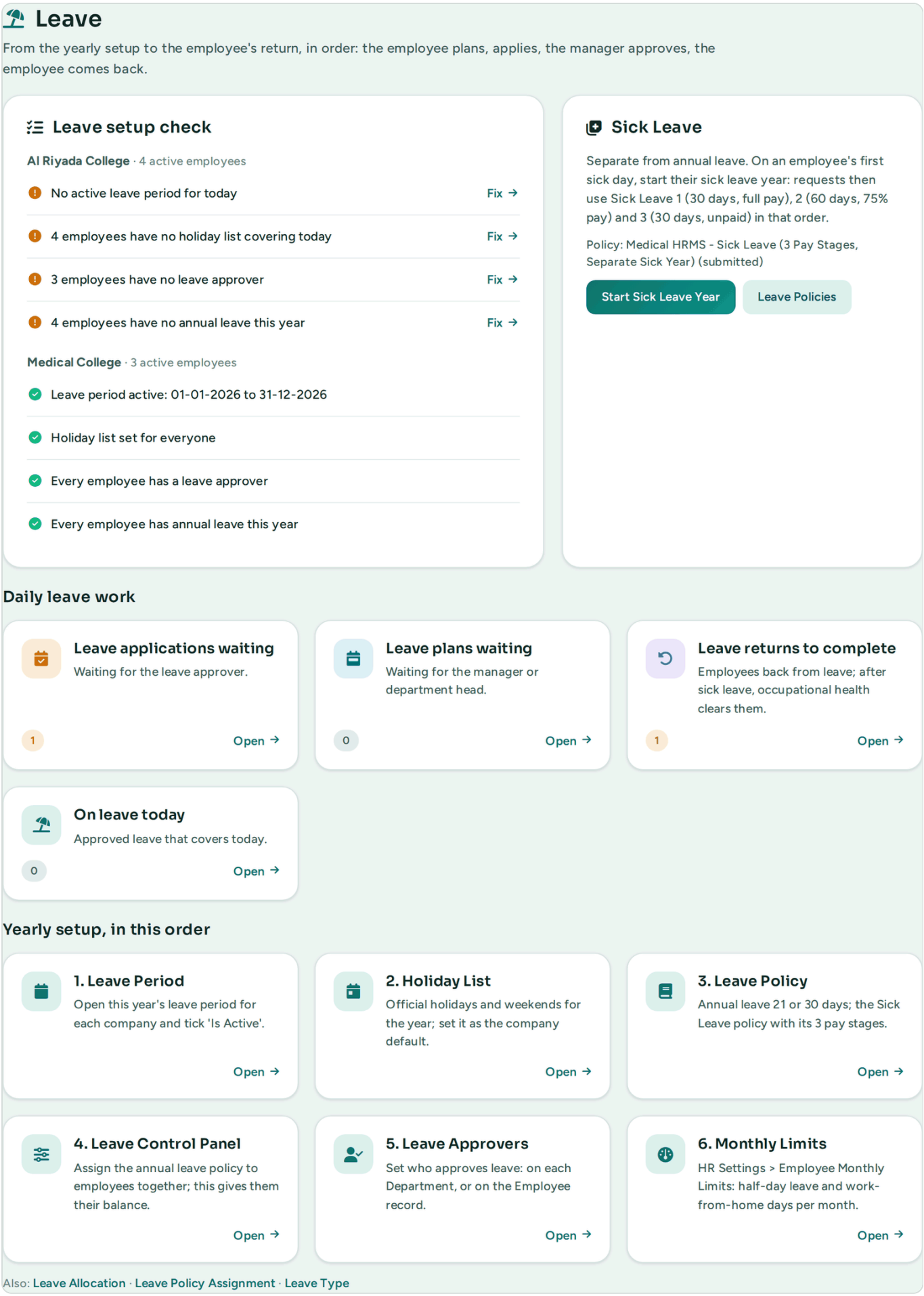



**Figure 2.** HR Portal → Leave. The setup check shows what is missing per company (orange, with a Fix link) and what is done (green). Below: daily leave work and the yearly setup steps.

1. **Leave Period**: create this year's leave period for the company and tick *Is Active*. Migrate creates an *inactive* starter period; check its dates and activate it.
2. **Holiday List**: enter the official holidays and weekends for the year and set it as the company's default holiday list.
3. **Leave Policy**: review the draft policies (annual leave 21 days, annual leave 30 days for five or more years of service, and the Sick Leave policy) and **submit** the ones the college uses.
4. **Leave Control Panel**: assign the annual leave policy to the employees together. This gives them their balance.
5. **Leave Approvers**: set who approves leave, on each Department (Leave Approvers table) or on each Employee.
6. **Monthly Limits**: HR Settings → Employee Monthly Limits (section 6.5).

> **Until this is done, employees cannot apply for leave**
> An employee with no annual leave assigned sees "No leave to apply for yet". Check the Leave setup check first when someone reports a leave problem.

## 6.4 Sick leave

Sick leave is separate from annual leave and follows the Saudi Labour Law: **Sick Leave 1** 30 days full pay, then **Sick Leave 2** 60 days at 75% pay, then **Sick Leave 3** 30 days unpaid, within one sick year that starts on the employee's first sick day.

1. Submit the policy *Medical HRMS - Sick Leave (3 Pay Stages, Separate Sick Year)*. Finance checks the 75% pay on a salary slip.
2. On an employee's first sick day, HR opens HR Portal → Leave → **Start Sick Leave Year**, picks the employee and the first sick day. This creates their Sick Leave 1 and 2 balances for one year; annual leave is not touched.
3. The employee then applies for sick leave from the Employee Portal; the stages are used in order and a request that crosses a stage must be split.

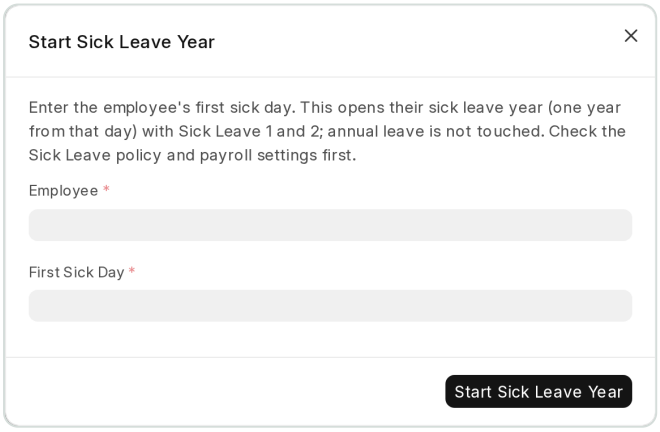



**Figure 3.** Start Sick Leave Year (HR Manager).

## 6.5 Monthly limits and academic presence

In **HR Settings**:

| Field | Meaning |
| --- | --- |
| **Enable Monthly Half-day Limit** | Limit half-day leave per calendar month. |
| **Leave Type Subject to Half-day Limit** | Which leave type the half-day limit applies to (usually annual leave). |
| **Half-day Requests per Calendar Month** | The limit. |
| **Enable Monthly Work-from-home Limit** | Limit remote-work days per calendar month. |
| **Work-from-home Working Days per Calendar Month** | The limit. |
| **Academic Holiday List** | Holiday list used by Academic Presence (days without classes). |
| **Suspends Regular Classes** | Event types that cancel classes on that day. |
| **Faculty After-hours Cut-off** | The time used by "Faculty After 4 PM". |

# 7 Jameah (Ministry) setup

Jameah is the Ministry's data system. Its agent reads the college's data from read-only database views in the Ministry's format. Medical HRMS creates and maintains these views; this section sets them up and checks them.

## 7.1 The 23 views

| Group | Views | One row per |
|---|---|---|
| **Employees (5)** | jameahEmployeePersonalInfo, …AcademicQualifications, … AcademicExperience, …PreviousExperience, … ProfessionalCertificates | Non-teaching employee of the reported company; table row |
| **Faculty (8)** | jameahFacultyPersonalInfo, …Research, …AcademicQualification, …AcademicExperience, …PreviousExperience, … ProfessionalCertificates, …Awards, …Attachments | Active Instructor; table row |
| **Students (10)** | jameahStudentPersonalInfo, …Companion, …ForeignAddress, … Relatives, …KSAAddress, …HighSchoolCertificate, …Contact, … Scholarship, …AcademicDetails, …Attachments | Student; guardian / certificate / document row |

- **Codes, not names**: lookup columns carry the official Ministry code. A value that is not a valid code for its list is sent empty, never guessed.
- Teaching staff are sent once, as **Faculty**, never also as Employees.
- Column names follow the Ministry document exactly (including its spellings `EnglishFourhtName` and `Summury`).

## 7.2 Jameah Settings

Open **HR Portal → Ministry (Jameah) → Jameah Settings**. Values left empty are sent empty. Academic Affairs confirms them with the Ministry.

| Setting | Meaning |
|---|---|
| **Educational Institution (UniversityId)** | RCHS is **P18** in the Ministry list. |
| **College (AdmissionCollegeId / CurrentCollegeId)** | The college code the Ministry assigned to RCHS. |
| **Country of Study** | Country of the campus. |
| **Study Location City** | Campus city, used when a student has no study city. |
| **Study Type for Regular / Bridging Students** | Study type sent for regular and for bridging students. |
| **GPA Type** | The grading scale of the GPA column. |
| **Program Duration and Unit** | Standard duration of the programme. |
| **File Base URL** | Prefix for attachment links, e.g. `https://erp.alriyada.edu.sa`. |
| **Company Reported as Employees** | Only this company's (non-teaching) employees are sent. |

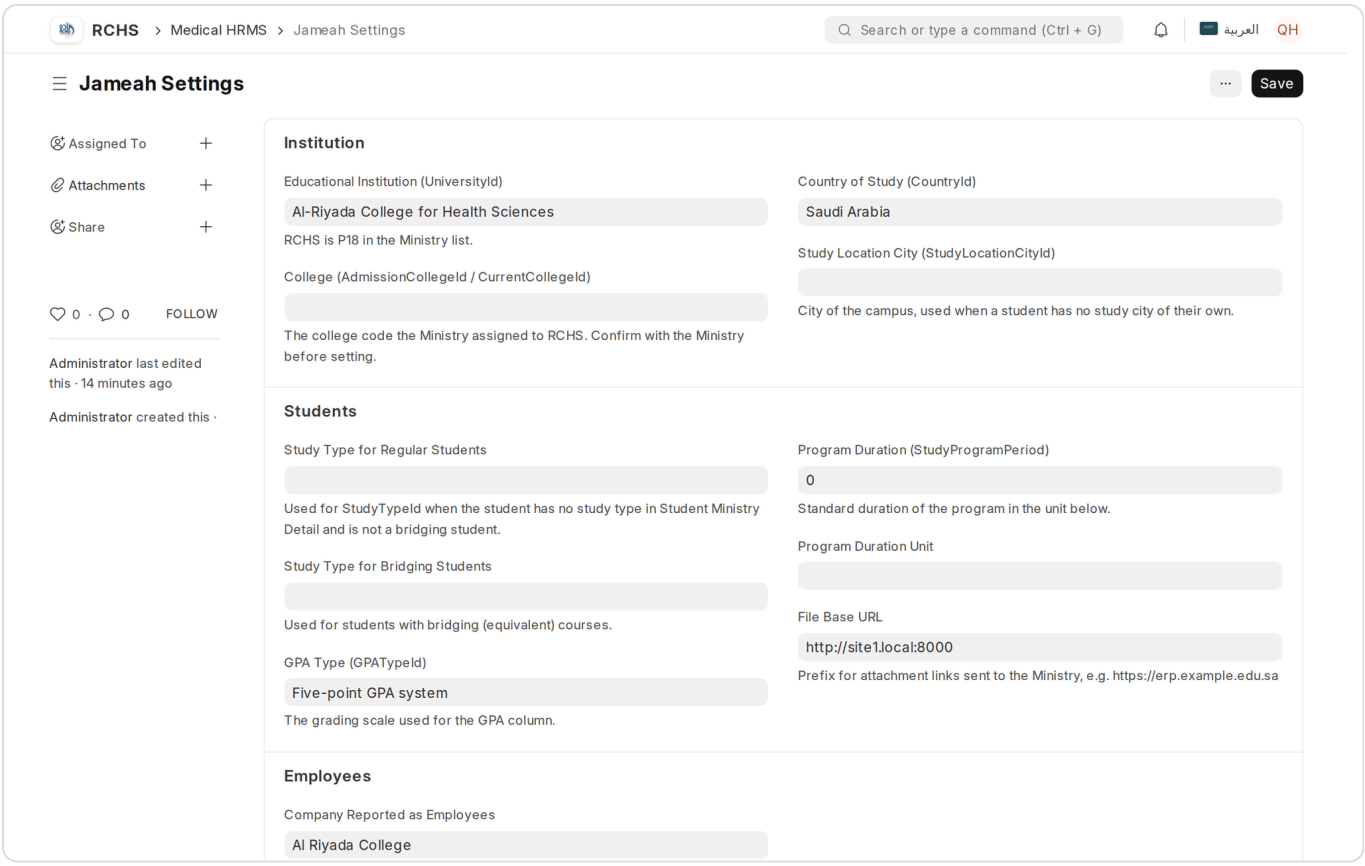



**Figure 4.** Jameah Settings.

## 7.3 Jameah Code Map

Some college lists keep their own codes (for example every city has code 90). The **Code Map** links each college value to the official Ministry code. Rows whose code and name agree with the official list are verified automatically; the others wait for a person.

1. Open **Ministry (Jameah) → Code Map**, filter *Status = Needs Review*.
2. Open a row, pick the Ministry code, set *Status = Verified*, save.
3. Until a row is verified, that value is sent empty.

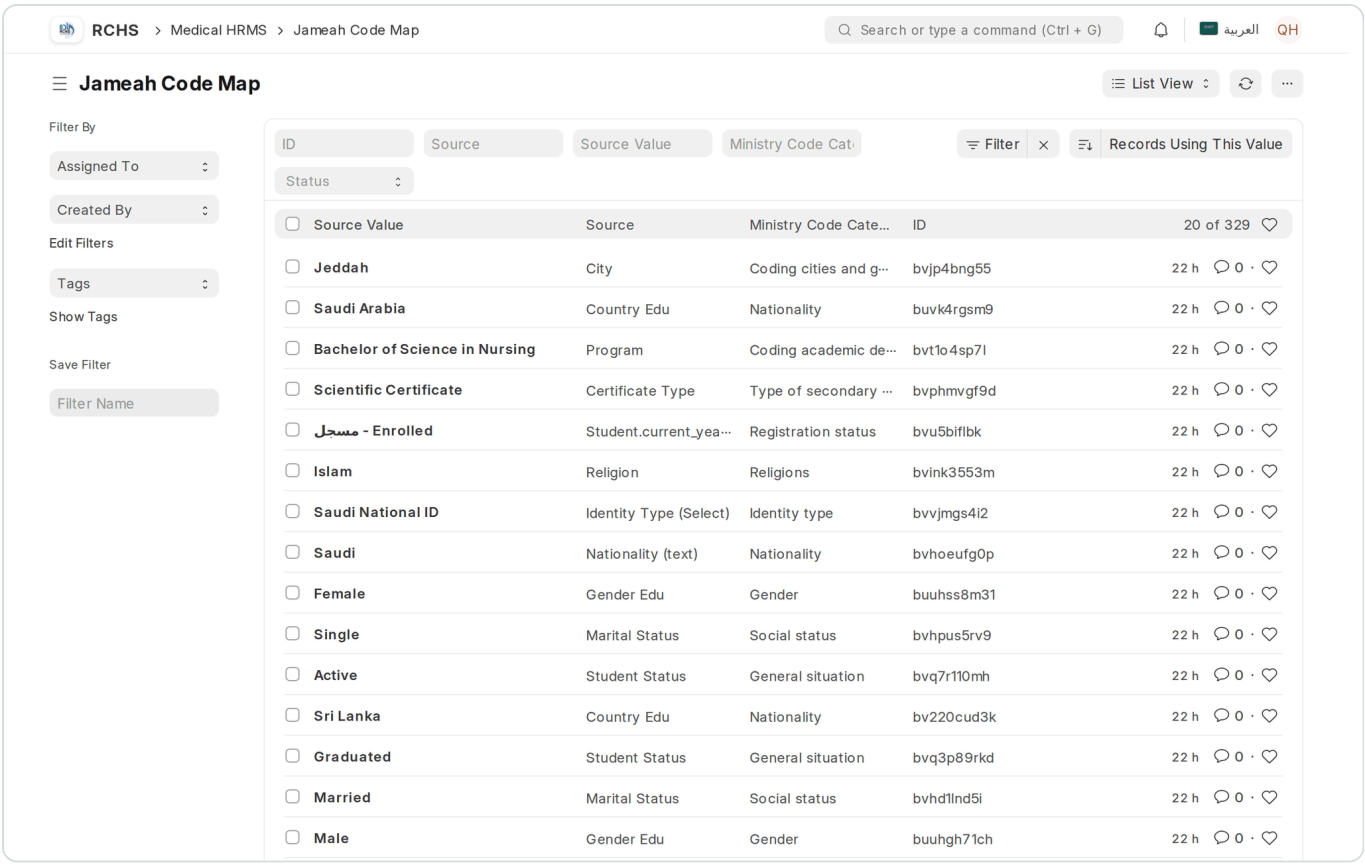



**Figure 5.** Jameah Code Map.

## 7.4 Jameah Data Check

The **Jameah Data Check** page shows what the views would send right now: invalid codes (must be 0), values waiting for a code, settings not filled, mandatory columns with gaps and problems in the source records. Use it after every data change.

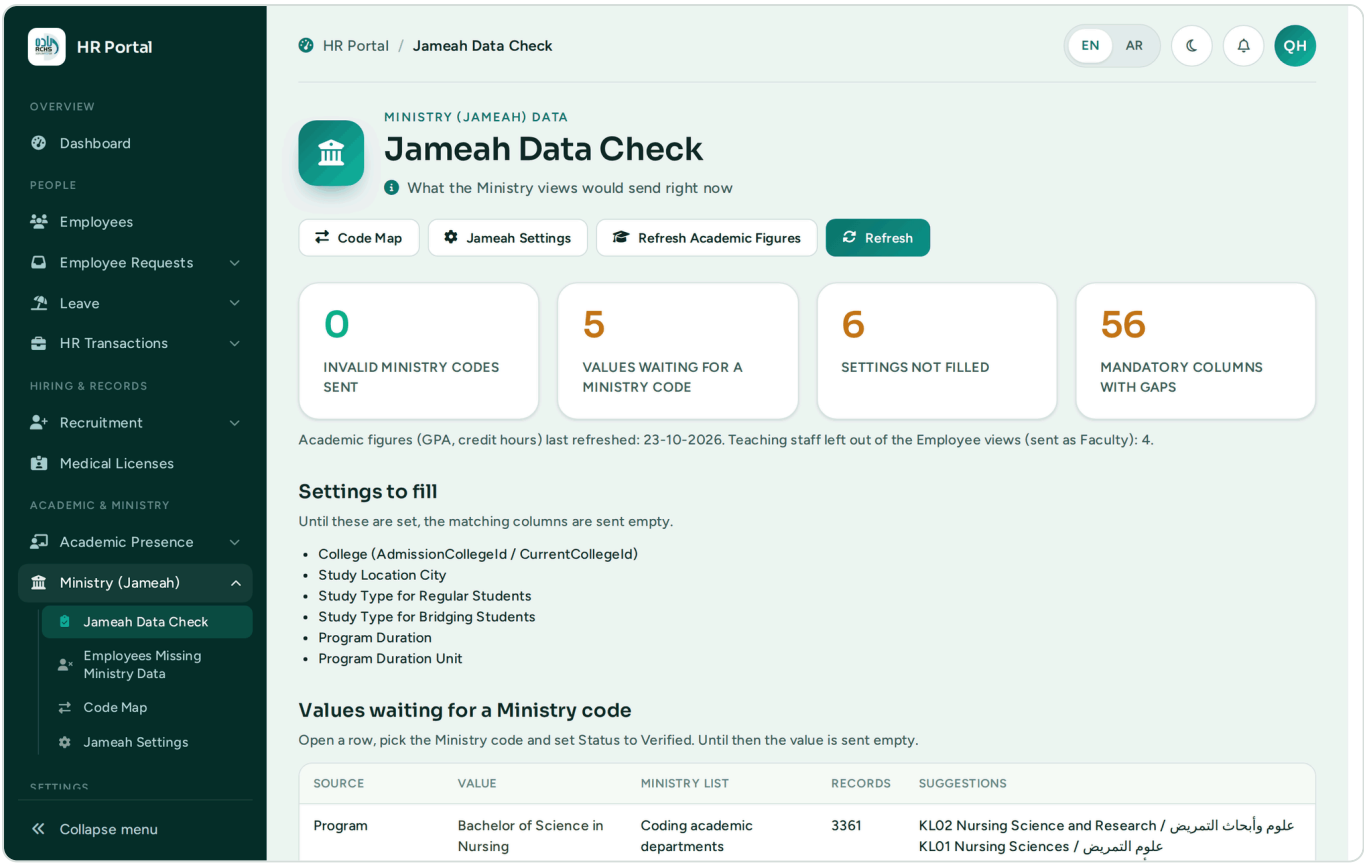



**Figure 6.** Jameah Data Check. Buttons at the top open the Code Map and Settings and refresh the academic figures.

## 7.5 Academic figures

GPA, credit hours and the latest enrolment come from a summary table rebuilt every hour (long queue, about 10 minutes), after every migrate, and with **Refresh Academic Figures** on the Data Check page. They equal the transcript (the college's official calculation). Only one rebuild runs at a time.

## 7.6 Verify every view

```
bench --site <site> execute medical_hrms.jameah.verify.run
```

Rebuilds every view independently in Python and compares it with the database view value by value. Every line must say **OK**; it also lists mandatory gaps and invalid codes per view.

## 7.7 Account for the Ministry agent

```
bench --site <site> execute medical_hrms.jameah.views.grant_statements
```

Prints the `CREATE USER` and `GRANT SELECT` statements for a read-only account that can read the 23 views and nothing else. Run them as the MariaDB administrator with a strong password, then give the Ministry the host, port, database, user and password.

## 7.8 Completing staff data

**Non-teaching staff** (HR Manager): open **Ministry (Jameah) → Employees Missing Ministry Data**. The report lists every employee the views send with each Ministry-mandatory value still empty.

1. **Download Data Sheet**: an Excel sheet with the current values; empty mandatory cells are red and Ministry lists are dropdowns.
2. HR fills it in.
3. **Check / Import Data Sheet → Check (nothing is saved)**: a result sheet lists each row's status.
4. When the check is clean, **Import**: values are saved through the normal Employee checks.

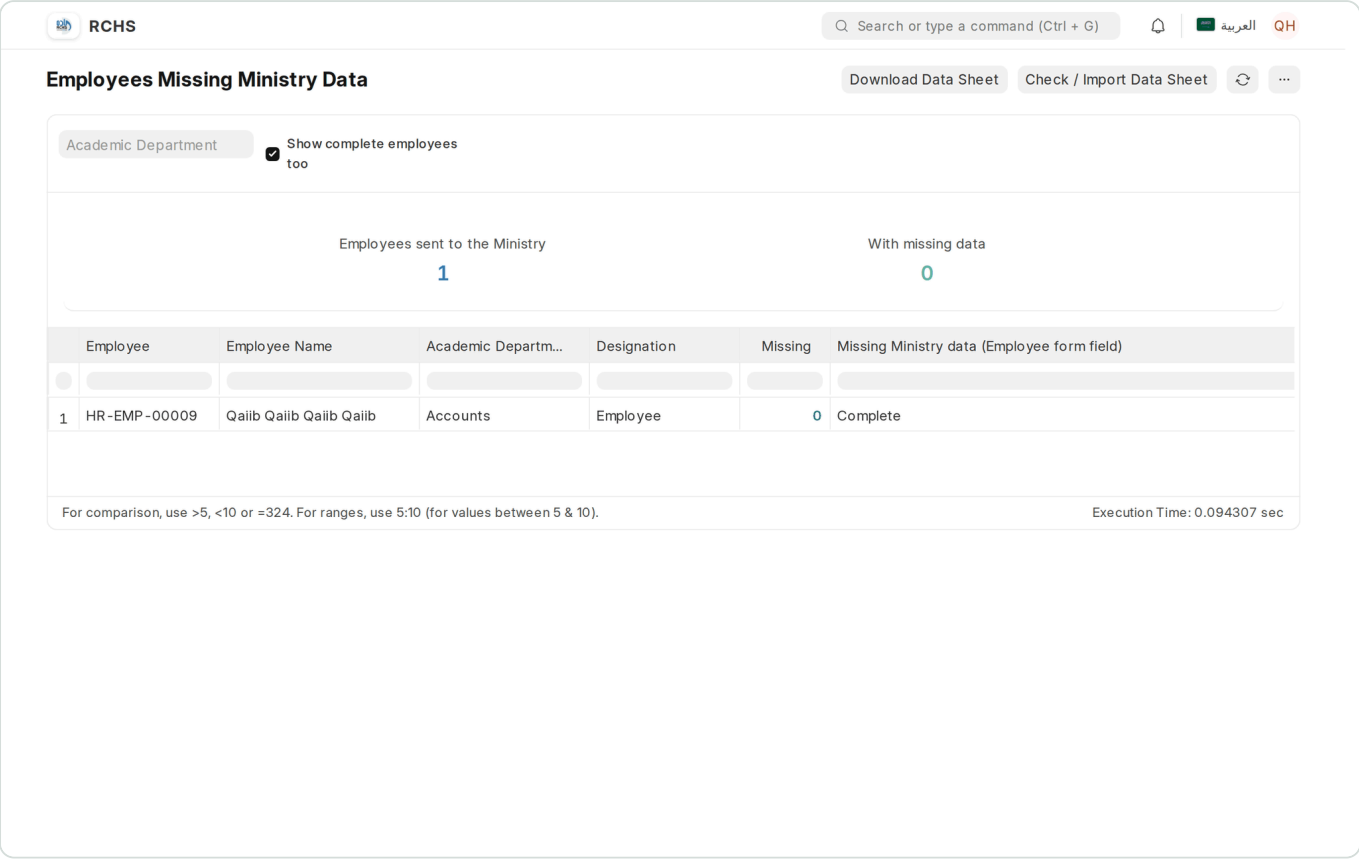



**Figure 7.** Employees Missing Ministry Data, with the data sheet buttons.

**Teaching staff** (existing Instructors without an Employee record): the administrator exports a sheet for HR, HR fills it, then it is checked and imported:

```
bench --site <site> execute medical_hrms.instructor_data_sheet.export_sheet
# HR fills the sheet and uploads it (Desk > File); check it (nothing is saved):
bench --site <site> execute medical_hrms.instructor_data_sheet.import_sheet --kwargs "{'file':
'/private/files/<name>.xlsx'}"
# when the check is clean, import for real:
bench --site <site> execute medical_hrms.instructor_data_sheet.import_sheet --kwargs "{'file':
'/private/files/<name>.xlsx', 'commit': 1}"
```

# 8 Testing the installation

> **Test sites only**
> The automatic browser tests create test logins with known passwords. They refuse to run unless the site is named in `QA_TEST_SITE`. Never run them on the live server.

| Test | Command (from `apps/medical_hrms`) | Passes when |
|---|---|---|
| **Every file** | `python3 tests/checks/files.py` | PROBLEMS: 0 |
| **Every hook, doctype and record** | `../../env/bin/python tests/e2e/fx.py tests/checks/site.py` | PROBLEMS: 0 |
| **Every server method, by role** | `python3 tests/checks/api_methods.py` (bench running, QA logins loaded) | TO REVIEW: 0 |
| **Unit tests** | `bench --site <test site> run-tests --app medical_hrms` (once per test site: `bench --site <test site> set-config allow_tests true`) | OK |
| **Ministry views** | `bench --site <site> execute medical_hrms.jameah.verify.run` | Every view OK |
| **Browser suite (95 checks)** | See `tests/e2e/README.md`: load the QA data, run `e2e_full.py`, remove the QA data | 95/95 passed |

Results of these checks go into the **Test Cases** workbook in `project_tracking/` (column *Last automated run*). The testing team then works through the same workbook by hand; see the User Guide, section "Testing the system".

> **Start the browser suite just after the hourly Jameah refresh**
> While the refresh runs, pages load slowly and checks can time out.

# 9 Troubleshooting

| Problem | What to do |
|---|---|
| **"App medical_hrms not in apps.txt"** | Put each app on its own line in `sites/apps.txt` (frappe, erpnext, payments, education, hrms, medical_hrms), then install and migrate again. |
| **An employee sees "No leave to apply for yet"** | They have no annual leave assigned: HR Portal → Leave setup check (section 6.3). |
| **"Failed to decrypt key ... password" in the Error Log** | Email Account *Notifications*: enter the password again and save (section 6.2). |
| **Pages slow for a few minutes every hour** | The Jameah academic summary is being rebuilt. Check the Scheduled Job Log (*summary.refresh*). On the live server, report it if it is noticeable. |
| **An HR user without the Employee role gets "Not permitted" on Employee Self Service** | Expected: that workspace is for employees. HR works from the HR Portal. |
| **Academic hire refused with "College E-mail Needed"** | Enter the official `@alriyada.edu.sa` address as Company Email in the Create Employee dialog. |
| **A Ministry column is empty** | Jameah Data Check: a setting not filled, a Code Map row not verified, or data missing on the record (report / data sheet). |
| **Logo missing on every page** | Upload `rchs-logo.png` again (Website Settings / File). |

# 10 Command reference

```
# install (first time)
bench get-app git@github.com:phpdevpro/edu-alriyada-hrm.git --branch develop
bench --site <site> install-app medical_hrms

# every update
cd apps/medical_hrms && git fetch origin && git checkout develop && git pull --ff-only origin
develop && cd ../..
bench --site <site> migrate
bench build --app medical_hrms
bench --site <site> clear-cache && bench restart

# checks
bench --site <site> doctor
bench --site <site> execute medical_hrms.jameah.verify.run

# Jameah
bench --site <site> execute medical_hrms.jameah.views.grant_statements      # read-only account
for the Ministry

# Ministry codes from a new Ministry workbook
bench --site <site> execute medical_hrms.import_ministry_excel_to_jameah_codes.execute --kwargs
"{'file_path':'/absolute/path/Ministry Code - (Translated).xlsx'}"

# teaching-staff data sheet
bench --site <site> execute medical_hrms.instructor_data_sheet.export_sheet
bench --site <site> execute medical_hrms.instructor_data_sheet.import_sheet --kwargs "{'file':
'/private/files/<name>.xlsx'}"
```

Documents in the repository: `README.md`, `DEVELOPER_SETUP.md`, `medical_hrms/jameah/README.md` and `VIEWS.md` (every Ministry column and its source), `project_tracking/` (Task Tracker, Test Cases, Checklist).